

Properties of integral

$$1) \int_a^b f(x) dx = - \int_b^a f(x) dx$$

integral go from a (lowest) to b (highest)
you can flip those around as needed but when
you do you must include a negative sign

$$\text{ex: } \int_{-2}^3 f(x) dx \Rightarrow - \int_3^{-2} f(x) dx$$

you can also flip it and make it go from high
to small but this is kind of useless unless you
want to cancel something just let it go from
small to big unless you need it to go in other
direction

$C = \text{constant}$

$$2) \int_a^b C dx \Rightarrow C \cdot (b-a)$$

always began with b and this only
works if there is only a constant not a
constant with equation

$$\text{ex: } \int_{-2}^1 5 dx \Rightarrow 5(1 - (-2)) = 15$$

$$3) \int_a^a f(x) dx = 0$$

there is no movement it went from point a to point a

$$4) \int_a^b f(x) \pm g(x) = \int_a^b f(x) \pm \int_a^b g(x)$$

$$\text{ex: } \int_1^2 x^2 \pm z^2 = \int_1^2 x^2 \pm \int_1^2 z^2$$

When you have two different terms you can break the integral into two and integrate the add or subtract

$$5) \int_a^b (c f(x)) dx = c \int_a^b f(x) dx$$

$$\text{ex: } \int_1^3 \frac{1}{2} x^2 = \frac{1}{2} \int_1^3 x^2$$

When there is a constant with another equation or anything else it can get pulled out this isn't the same as the other one the other one only has to have the term by itself while this works with equation

6) $\int_a^c f(x) dx + \int_c^b f(x) dx$

ex: given = $\int_c^b f(x) = 1$ $\int_c^a f(x) = 5$

Find $\int_b^a f(x) = 4$

$$\int_c^b f(x) + \int_c^a f(x)$$

$$= \int_b^c f(x) + \int_c^a f(x)$$

$$= (1) + 5 = 4$$